Name
Date
Course

Applications of Algebra (Part I): Using Systems of Equations

① There are red bottle caps and blue bottle caps in a bag. There is a total of 56 bottle caps in the bag. There are twelve more red caps than blue caps. How many caps of each color are in the bag?

R = # of red caps
B = # of blue caps

$$\begin{cases} R + B = 56 \\ R = 12 + B \end{cases}$$

$$12 + B + B = 56$$

$$12 + 2B = 56$$

$$\boxed{B = 22}$$

$$R = 12 + 22$$

$$\boxed{R = 34}$$

② There are three times as many horses as mules. The difference between the number of horses and mules is 12. How many of each type of animal are there?

H = # of horses
M = # of mules

$$\begin{cases} H - M = 12 \\ H = 3M \end{cases}$$

$$3M - M = 12$$

$$2M = 12$$

$$\boxed{M = 6}$$

$$H = 3(6)$$

$$\boxed{H = 18}$$

③ Billy has 21 cards in his pocket. Each one is either a baseball card or a football card. There are five fewer football cards than baseball cards. How many of each type of card does he have in his pocket?

B = # of baseball cards
F = # of football cards

$$\begin{cases} B + F = 21 \\ F = B - 5 \end{cases}$$

$$B + B - 5 = 21$$

$$2B - 5 = 21$$

$$\boxed{B = 13}$$

$$F = 13 - 5$$

$$\boxed{F = 8}$$

④ The difference between the number of cookies and brownies at a bake sale is 40. If there are twice as many cookies as brownies, how many of each dessert are there?

C = # of cookies
B = # of brownies

$$\begin{cases} C - B = 40 \\ C = 2B \end{cases}$$

$$2B - B = 40$$

$$\boxed{B = 40}$$

$$C = 2(40)$$

$$\boxed{C = 80}$$

⑤ There are 18 people at a government meeting. Every person at the meeting is either a Democrat or a Republican. If the number of Republicans is two more than three times the number of Democrats, how many Democrats are at the meeting?

D = # of Democrats
R = # of Republicans

$$\begin{cases} D+R=18 \\ R=2+3D \end{cases}$$

$$D+2+3D=18$$

$$4D+2=18$$

$$\boxed{D=4}$$

⑥ There are 34 students in a classroom. The number of boys is eight less than two times the number of girls. How many girls are there?

G = # of girls
B = # of boys

$$\begin{cases} G+B=34 \\ B=2G-8 \end{cases}$$

$$G+2G-8=34$$

$$3G-8=34$$

$$\boxed{G=14}$$

Money Problems

⑦ A guy at a theme park sells cotton candy and salted pretzels. He sells cotton candy for \$6 each and pretzels for \$7 each. In one day, the man made 38 sales totaling \$235. How many of each item did he sell?

C = # of cotton candy sticks
P = # of pretzels

6C = Amount of money he collected from the cotton candy.

7P = Amount of money he collected from pretzels.

$$\begin{cases} C+P=38 \\ 6C+7P=235 \end{cases}$$

$$\begin{cases} -6(C+P)=-6(38) \\ 6C+7P=235 \end{cases}$$

$$-6C-6P=-228$$
$$6C+7P=235$$

$$\boxed{P=7}$$

$$C+7=38$$
$$\boxed{C=31}$$

⑧ At a play, the price of admission for an adult is \$25, whereas the price of admission for a child is \$15. The theater sold 86 tickets and received \$1,910. How many of each type of ticket did the theater sell?

A = # of adult tickets
C = # of child tickets

$$\begin{cases} A+C=86 \\ 25A+15C=1{,}910 \end{cases}$$

$$\begin{cases} -15A-15C=-1{,}290 \\ 25A+15C=1{,}910 \end{cases}$$

$$10A = 620$$

$$\boxed{A=62}$$

$$62+C=86$$

$$\boxed{C=24}$$

⑨ A restaurant sells small hamburgers for \$4 each and large hamburgers for \$5 each. If the store sold 46 hamburgers for \$219, how many hamburgers of each size were sold?

S = # of small hamburgers
L = # of large hamburgers

$$\begin{cases} S+L=46 \\ 4S+5L=219 \end{cases}$$

$$\begin{cases} -4S-4L=-184 \\ 4S+5L=219 \end{cases}$$

$$\boxed{L=35}$$

$$S+35=46$$

$$\boxed{S=11}$$

⑩ A vendor at a parade sells whistles and hats. He sells whistles for \$2 each and hats for \$3 each. If the vendor makes 32 sales totaling \$77, how many of each item did he sell?

W = # of whistles
H = # of hats

$$\begin{cases} W+H=32 \\ 2W+3H=77 \end{cases}$$

$$\begin{cases} -2W-2H=-64 \\ 2W+3H=77 \end{cases}$$

$$\boxed{H=13}$$

$$W+13=32$$

$$\boxed{W=19}$$

⑪ Dorian bought three tacos and two quesadillas for \$14. Lisa bought four tacos and one quesadilla for \$12. What was the price per item?

T = Price per taco
Q = Price per quesadilla

$$\begin{cases} 3T+2Q=14 \\ 4T+Q=12 \end{cases}$$

$$\begin{cases} 3T+2Q=14 \\ -8T-2Q=-24 \end{cases}$$

$$-5T=-10$$

$$\boxed{T=\$2/\text{taco}}$$

$$4(2)+Q=12$$

$$8+Q=12$$

$$\boxed{Q=\$4/\text{quesadilla}}$$

⑫ A school bought four soccer balls and three footballs for a total of \$135. A different school bought five soccer balls and two footballs for a total of \$132. What was the price per item?

S = Price per soccer ball
F = Price per football

$$\begin{cases} 4S+3F=135 \\ 5S+2F=132 \end{cases}$$

$$\begin{cases} -8S-6F=-270 \\ 15S+6F=396 \end{cases}$$

$$7S=126$$

$$\boxed{S=\$18/\text{ball}}$$

$$5(18)+2F=132$$

$$90+2F=132$$

$$\boxed{F=\$21/\text{ball}}$$

13) One day a store sold five Mega Mice games and six Earth Man games for a total of \$140. The next day the store sold eight Mega Mice games and two Earth Man games for a total of \$110. What was the price per game?

M = Price per Mega Mice game
E = Price per Earth Man game

$$\begin{cases} 5M+6E=140 \\ 8M+2E=110 \end{cases}$$

$$\begin{cases} 5M+6E=140 \\ -24M-6E=-330 \end{cases}$$

$$-19M = -190$$

$$\boxed{M = \$10/\text{game}}$$

$$8(10)+2E=110$$

$$80+2E=110$$

$$\boxed{E = \$15/\text{game}}$$

14) Robert sold candy to raise money for his baseball team. One day he sold eleven Delicious Bars and five Sour Sticks for \$43. The next day he sold two Delicious Bars and three Sour Sticks for \$12. What price did he charge per item?

D = Price per Delicious Bar
S = Price per Sour stick

$$\begin{cases} 11D+5S=43 \\ 2D+3S=12 \end{cases}$$

$$\begin{cases} -33D-15S=-129 \\ 10D+15S=60 \end{cases}$$

$$-23D=-69$$

$$\boxed{D = \$3/\text{bar}}$$

$$2(3)+3S=12$$

$$6+3S=12$$

$$\boxed{S = \$2/\text{stick}}$$

Decimal Problems (You may use a calculator)

15) Donny bought stickers and pins at his school store. Stickers were \$1.19 each and pins were \$3.13 each and he paid a total of \$19.22. If he bought eight items, how many stickers did he buy and how many pins did he buy?

S = # of stickers
P = # of pins

$$\begin{cases} S+P=8 \\ 1.19S+3.13P=19.22 \end{cases}$$

$$\begin{cases} -1.19S-1.19P=-9.52 \\ 1.19S+3.13P=19.22 \end{cases}$$

$$1.94P=9.7$$

$$\boxed{P=5}$$

$$S+5=8$$

$$\boxed{S=3}$$

16) Larry bought beef jerky and potatoes for a total of \$21.54. If beef jerky was \$6.15 per pound and potatoes were \$3.08 per pound, and he bought a total of five pounds of these items, how many pounds of each item did he buy?

J = # of pounds jerky
P = # of pounds potatoes

$$\begin{cases} J+P=5 \\ 6.15J+3.08P=21.54 \end{cases}$$

$$\begin{cases} -3.08J-3.08P=-15.4 \\ 6.15J+3.08P=21.54 \end{cases}$$

$$3.07J=6.14$$

$$\boxed{J = 2 \text{ lbs of jerky}}$$

$$2+P=5$$

$$\boxed{P = 3 \text{ lbs Potatoes}}$$

⑰ Natalie purchased eight boxes of tissue and five rolls of paper towels for a total of \$13.92. Tom purchased four boxes of tissue and two rolls of paper towels for a total of \$6.52. Assuming they bought the same brand and type of each item, what was the cost per box of tissue and the cost per roll of paper towels?

B = Price per box of tissue
R = Price per roll of paper towels

$$\begin{cases} 8B+5R=13.92 \\ 4B+2R=6.52 \end{cases}$$

$$\begin{cases} 8B+5R=13.92 \\ -8B-4R=-13.04 \end{cases}$$

$$\boxed{R=\$0.88/\text{roll}}$$

$$4B+2(0.88)=6.52$$
$$4B+1.76=6.52$$
$$\boxed{B=\$1.19/\text{box}}$$

⑱ Lily bought five 10" by 10" canvases and three large diameter paint brushes. She paid \$45.55. Jorge bought four 10" by 10" canvases and seven large diameter brushes for a total of \$73.70. What is the cost per brush and the cost per canvas?

B = Price per brush
C = Price per canvas

$$\begin{cases} 5C+3B=45.55 \\ 4C+7B=73.70 \end{cases}$$

$$\begin{cases} -20C-12B=-182.2 \\ 20C+35B=368.5 \end{cases}$$

$$23B=186.3$$
$$\boxed{B=\$8.10/\text{brush}}$$

$$4C+7(8.1)=73.7$$
$$4C+56.7=73.7$$
$$\boxed{C=\$4.25/\text{canvas}}$$

Coin Problems

⑲ A change machine contains nickels and quarters. There are a total of 36 coins. The total amount of money in the machine is \$5.60. How many of each type of coin are in the machine?

N = # of nickels
Q = # of quarters

$$\begin{cases} N+Q=36 \\ 5N+25Q=560 \end{cases}$$

$$\begin{cases} -5N-5Q=-180 \\ 5N+25Q=560 \end{cases}$$

$$20Q=380$$
$$\boxed{Q=19}$$

$$N+19=36$$
$$\boxed{N=17}$$

⑳ There are nickels and dimes in a bag. The total number of nickels and dimes is 15. The total amount of money in the bag is \$1.05. How many of each coin are in the bag?

N = # of nickels
D = # of dimes

$$\begin{cases} N+D=15 \\ 5N+10D=105 \end{cases}$$

$$\begin{cases} -5N-5D=-75 \\ 5N+10D=105 \end{cases}$$

$$5D=30$$
$$\boxed{D=6}$$

$$N+6=15$$
$$\boxed{N=9}$$

(21) Cynthia has only quarters and dimes in her purse. There are a total of 16 coins. If the coins are worth \$2.05, how many of each type of coin does she have in her purse?

$Q = $ # of quarters
$D = $ # of dimes

$$\begin{cases} Q + D = 16 \\ 25Q + 10D = 205 \end{cases}$$

$$\begin{cases} -10Q - 10D = -160 \\ 25Q + 10D = 205 \end{cases}$$

$$15Q = 45$$

$$\boxed{Q = 3}$$

$$3 + D = 16$$

$$\boxed{D = 13}$$

(22) Using a metal detector, Thomas found 11 coins at the beach. The total value of the coins was \$1.95. If he found only nickels and quarters, how many of each type of coin did he find?

$N = $ # of nickels
$Q = $ # of quarters

$$\begin{cases} N + Q = 11 \\ 5N + 25Q = 195 \end{cases}$$

$$\begin{cases} -5N - 5Q = -55 \\ 5N + 25Q = 195 \end{cases}$$

$$20Q = 140$$

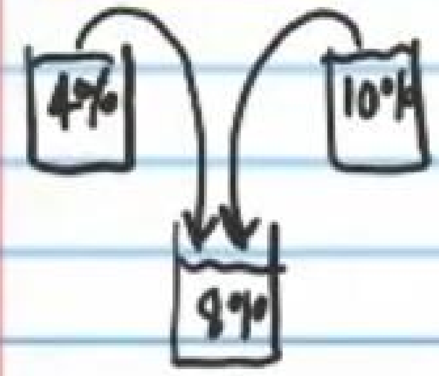
$$\boxed{Q = 7}$$

$$N + 7 = 11$$

$$\boxed{N = 4}$$

Mixing Problems

(23) Dr. Peterson wants to prepare 12 ounces of an 8% saline solution (a mixture of salt and water), but in her lab she has access to only 4% saline solution and 10% saline solution. How many ounces of each solution must she mix to produce her desired result?



$V_4 = $ Volume of 4% solution (ounces).
$V_{10} = $ Volume of 10% solution (ounces).

$$\begin{cases} V_4 + V_{10} = 12 \\ 0.04V_4 + 0.1V_{10} = 0.08(12) \end{cases}$$

$0.04V_4 = $ salt

$0.10V_{10} = $ salt

$$\begin{cases} V_4 + V_{10} = 12 \\ 0.04V_4 + 0.1V_{10} = 0.96 \end{cases}$$

$$\begin{cases} -0.1V_4 - 0.1V_{10} = -1.2 \\ 0.04V_4 + 0.1V_{10} = 0.96 \end{cases}$$

$$-0.06V_4 = -0.24$$

$$\boxed{V_{10} = 8\text{ oz}} \qquad \boxed{V_4 = 4\text{ oz}}$$

(24) Three karat gold is 12.5% gold. Twelve karat gold is 50% gold. If a metallurgist has only three karat gold and 12 karat gold, how many grams of each purity should he melt and mix to create twelve grams of six karat gold (25% gold)?

$M_{12.5} = $ Mass of 12.5% substance (grams)
$M_{50} = $ Mass of 50% substance (grams)

$$\begin{cases} M_{12.5} + M_{50} = 12 \\ 0.125M_{12.5} + 0.5M_{50} = 0.25(12) \end{cases}$$

$$\begin{cases} M_{12.5} + M_{50} = 12 \\ 0.125M_{12.5} + 0.5M_{50} = 3 \end{cases}$$

$$\begin{cases} -0.5M_{12.5} - 0.5M_{50} = -6 \\ 0.125M_{12.5} + 0.5M_{50} = 3 \end{cases}$$

$$-0.375M_{12.5} = -3$$

$$\boxed{M_{12.5} = 8\text{ g}}$$

$$\boxed{M_{50} = 4\text{ g}}$$

(25) Vivien wants to make 36 milliliters of a 6% hydrochloric acid solution (water and acid). In her laboratory she has 3% and 12% solutions. How many milliliters of each of these solutions should she mix to produce her desired result?

V_3 = Volume of 3% solution (ml).
V_{12} = Volume of 12% solution (ml).

$$\begin{cases} V_3 + V_{12} = 36 \\ 0.03V_3 + 0.12V_{12} = 0.06(36) \end{cases}$$

$$\begin{cases} V_3 + V_{12} = 36 \\ 0.03V_3 + 0.12V_{12} = 2.16 \end{cases}$$

$$\begin{cases} -0.03V_3 - 0.03V_{12} = -1.08 \\ 0.03V_3 + 0.12V_{12} = 2.16 \end{cases}$$

$$0.09V_{12} = 1.08$$

$$\boxed{V_{12} = 12 \text{ ml}}$$

$$V_3 + 12 = 36$$

$$\boxed{V_3 = 24 \text{ ml}}$$

(26) A builder has two types of concrete. One type is 10% cement and the other type is 14% cement. How many kilograms of each type of concrete should the builder mix to create 32 kilograms of a concrete that is 13.4% cement?

M_{10} = Mass of 10% substance (kg)
M_{14} = Mass of 14% substance (kg)

$$\begin{cases} M_{10} + M_{14} = 32 \\ 0.1M_{10} + 0.14M_{14} = 0.134(32) \end{cases}$$

$$\begin{cases} M_{10} + M_{14} = 32 \\ 0.1M_{10} + 0.14M_{14} = 4.288 \end{cases}$$

$$\begin{cases} -0.1M_{10} - 0.1M_{14} = -3.2 \\ 0.1M_{10} + 0.14M_{14} = 4.288 \end{cases}$$

$$0.04M_{14} = 1.088$$

$$\boxed{M_{14} = 27.2 \text{ kg}}$$

$$M_{10} + 27.2 = 32$$

$$\boxed{M_{10} = 4.8 \text{ kg}}$$

Combining Money Problems with Mixing Problems

(27) Lemonade concentrate costs 50¢ per ounce. Water costs 10¢ per ounce. How many ounces of each ingredient are necessary to make 16 ounces of lemonade that costs 27¢ per ounce?

V_w = Volume of water.
V_c = Volume of concentrate.

$10V_w$ = Money paid for water.
$50V_c$ = Money paid for concentrate.

$$\begin{cases} V_w + V_c = 16 \\ 10V_w + 50V_c = 27(16) \end{cases}$$

$$\begin{cases} V_w + V_c = 16 \\ 10V_w + 50V_c = 432 \end{cases}$$

$$\begin{cases} -10V_w - 10V_c = -160 \\ 10V_w + 50V_c = 432 \end{cases}$$

$$40V_c = 272$$

$$\boxed{V_c = 6.8 \text{ oz}}$$

$$V_w + 6.8 = 16$$

$$\boxed{V_w = 9.2 \text{ oz}}$$

(28) Metal A costs \$28.25 per kilogram, while metal B costs \$25.50 per kilogram. If the metals are mixed to form a 90 kilogram alloy, how many kilograms of each metal are necessary to make the alloy cost \$27.70 per kilogram?

A = Mass of Metal A (kg).
B = Mass of Metal B (kg).

$$\begin{cases} A + B = 90 \\ 28.25A + 25.5B = 27.7(90) \end{cases}$$

$$\begin{cases} A + B = 90 \\ 28.25A + 25.5B = 2{,}493 \end{cases}$$

$$\begin{cases} -25.5A - 25.5B = -2295 \\ 28.25A + 25.5B = 2{,}493 \end{cases}$$

$$2.75A = 198$$

$$\boxed{A = 72 \text{ kg}}$$

$$72 + B = 90$$

$$\boxed{B = 18 \text{ kg}}$$

(29) Lana wants to sell trail mix that is a mixture of candy and peanuts. If the peanuts cost \$4 per pound and the candy costs \$15 per pound, how many pounds of each ingredient must she mix to create 20 pounds of trail mix that costs \$9 per pound?

P = # of pound peanuts.
C = # of pounds candy.

$$\begin{cases} P+C=20 \\ 4P+15C=9(20) \end{cases}$$

$$\begin{cases} P+C=20 \\ 4P+15C=180 \end{cases}$$

$$\begin{cases} -4P-4C=-80 \\ 4P+15C=180 \end{cases}$$

$$11C=100$$

$$\boxed{C=9.\overline{09}\text{ lbs}}$$

$$P+9.\overline{09}=20$$

$$\boxed{P=10.\overline{90}\text{ lbs}}$$

(30) A gardening store sells soil for \$2.75 per pound and a fertilizer for \$11.75 per pound. How many pounds of each ingredient are necessary to produce 20 pounds of mixture that costs \$9.05 per pound?

S = # of pounds soil.
F = # of pounds fertilizer.

$$\begin{cases} S+F=20 \\ 2.75S+11.75F=9.05(20) \end{cases}$$

$$\begin{cases} S+F=20 \\ 2.75S+11.75F=181 \end{cases}$$

$$\begin{cases} -2.75S-2.75F=-55 \\ 2.75S+11.75F=181 \end{cases}$$

$$9F=126$$

$$\boxed{F=14\text{ lbs}}$$

$$S+14=20$$

$$\boxed{S=6\text{ lbs}}$$